



SECJ1023
PROGRAMING TECHNIQUE II
SECTION 2
PROJECT

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PART 1: INTRODUCTION

1.1 Synopsis Project

The purpose of this proposal is to suggest the development of an OOP-based Income Tax Calculator System that utilizes principles composition. This system will be designed to help individuals or business estimate the amount of income tax they owe to the government based on their income and various tax-related factors.

The Income Tax Calculator System is a software solution provides a convenient and efficient way for individuals and businesses to estimate their tax liabilities and plan their finances accordingly. In user menu, the system will have a user-friendly interface where individuals or businesses can input their financial information. The system will ask user to key-in various amounts that will affect the final amount of income tax. Meanwhile, admin will be able to update the latest income tax info since the rate and tax deducted might be different every year. With the use of C++ programming, Income Tax Calculator System helps users better understand and manage their tax obligations by being accurate, efficient and user-friendly.

1.2 Objective of The Project

1. **Accurate Tax Calculation:** To develop an accurate and reliable system that calculates income tax based on the latest tax rates and regulations, ensuring compliance with current tax laws.
2. **User-Friendly Interface:** To design an intuitive and easy-to-use interface that allows individuals and businesses to input their financial data efficiently, enhancing the user experience and reducing the likelihood of errors.
3. **Dynamic Tax Information Management:** To enable administrative functionality for updating and maintaining the latest tax rates, deductions, and other relevant tax information, ensuring the system remains current and relevant.
4. **Comprehensive Reporting:** To incorporate reporting features that allow users to generate detailed tax summaries, reports, and forecasts, facilitating better financial decision-making.

PART 2: SYSTEM ANALYSIS AND DESIGN

2.1 System Requirements

Use case diagram

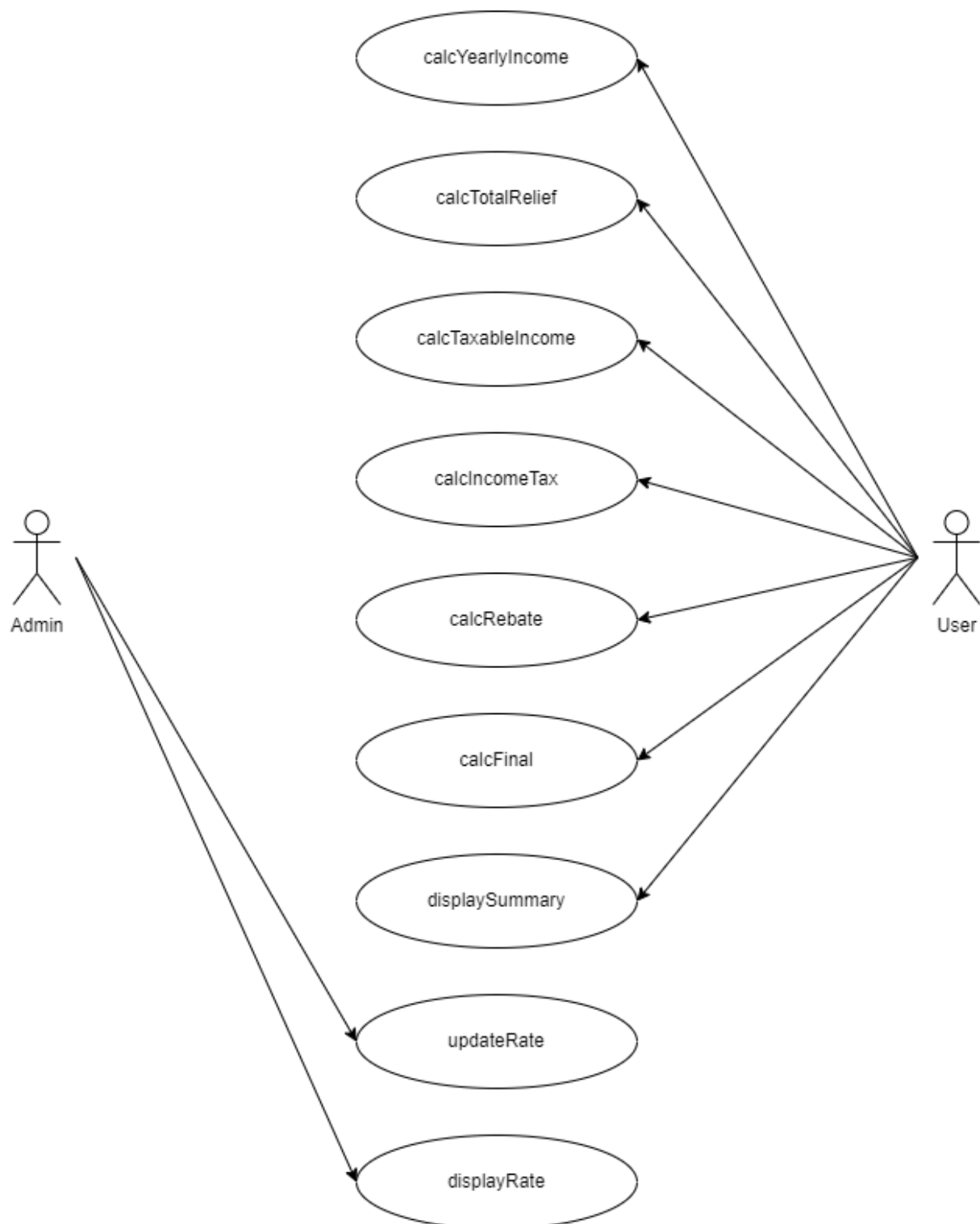


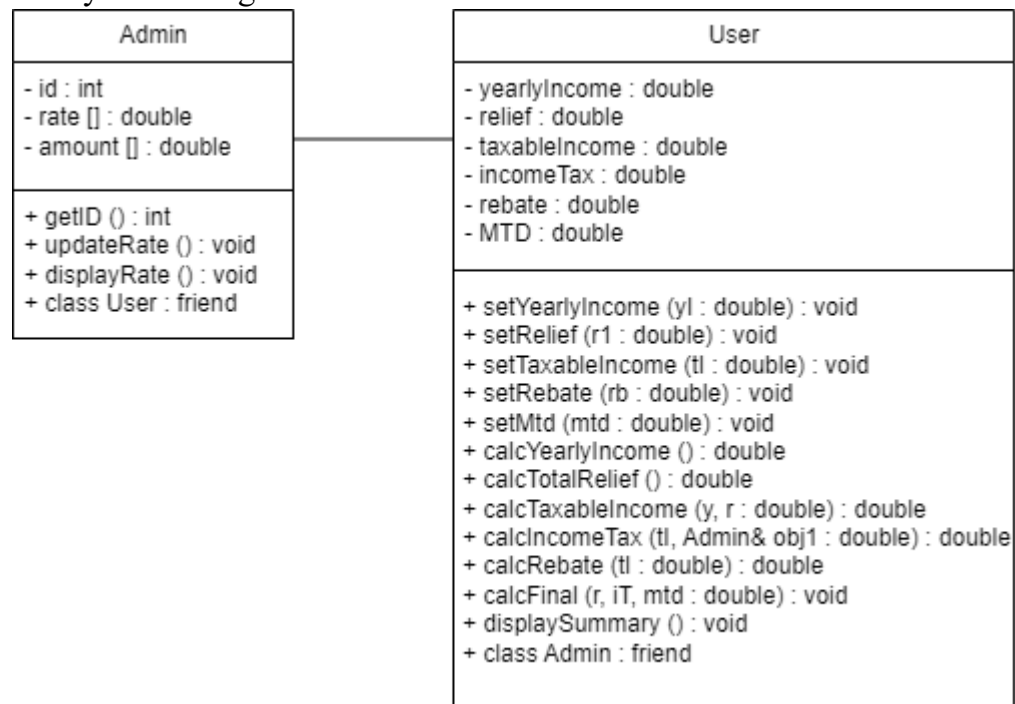
Figure 1 : Use Case Diagram for Tax Calculation System

Detail Description for Each Use Cases

The system has 9 main use cases.

Use Case	Purpose
calcYearlyIncome	The passive and active income are keyed into the system by the user to get the total yearly income
calcTotalRelief	To calculate the total tax relief that could be rebated by the user
calcTaxableIncome	Calculating the final taxable income to be taxed after deducting the total relief
calcIncomeTax	To calculate the income tax should be paid by user
calcRebate	Determine whether the user able to get a tax rebate (taxable income < RM35000 / zakat) for deduction in income tax
calcFinal	Calculate and determine the final income tax needed to pay or returned by LHDN
displaySummary	Display all the details and the price for yearly income, total relief, taxable income, income tax, etc
updateRate	To update the newest yearly income tax rate into the system
displayRate	Final income tax will be displayed after calculation

2.2 System Design



PART 3: System Prototype

```
----- Welcome to Income Tax Calculation System -----  
  
1. Admin  
2. User  
3. Exit  
  
Please enter your choice (1/2/3): |
```

Screen 1: Main menu

Screen 1: The user must insert the integer value in the range 1 – 3. If the user enters other number, the system will prompt user to enter again the integer value for choice.

```
----- Welcome to the Admin Page -----  
  
Please enter your admin ID : 001  
Incorrect ID...  
  
Please enter your admin ID : 007  
Welcome!
```

Screen 2: Admin ID checking

Screen 2: When enter value 1, it will proceed to the admin page and able to change the rate of the system. Admin are required to enter correct id to make sure the secure of admin system. If the id was incorrect, it will prompt user to enter again the id.

```

----- Current Rate -----
Rate 1 :0.000
Rate 2 :0.010
Rate 3 :0.030
Rate 4 :0.080
Rate 5 :0.140
Rate 6 :0.210
Rate 7 :0.240
Rate 8 :0.245
Rate 9 :0.250
Rate 10 :0.260
Rate 11 :0.280
Rate 12 :0.300

-----

Please enter the NEW income tax rate

Rate 1 : 0.01
Rate 2 : 0.015
Rate 3 : 0.02
Rate 4 : 0.025
Rate 5 : 0.03
Rate 6 : 0.035
Rate 7 : 0.04
Rate 8 : 0.045
Rate 9 : 0.05
Rate 10 : 0.055
Rate 11 : 0.06
Rate 12 : 0.065

-----

Tax rate renew successfully!

```

Screen 3: Changing Rate

Screen 3: The current rate will be display after enter the admin id and prompt user to update the new rate for income tax calculation. When the rate updated, it shows tax rate renew successfully.

```

Do you want to test the income tax calculation system ?
1. Yes
2. No

Enter your choice :|

Do you want to test the income tax calculation system ?
1. Yes
2. No

Enter your choice :1

----- Welcome to the User Page -----

Please enter the details of your ANNUAL INCOME :
Active income = RM

```

```
Do you want to test the income tax calculation system ?
1. Yes
2. No

Enter your choice :2
Logging out from Admin Page....
```

Screen 4: Choice of test

Screen 4: After updated the rate of calculation, the system will ask to test the income tax calculation system or not. If yes, admin will be brought to the testing system but if no, admin will be log out from the page.

```
----- Welcome to the User Page -----

Please enter the details of your ANNUAL INCOME :
Active income  = RM40000
Passive income = RM900

Please enter the details of TAX RELIEF :
Amount of individual & dependent relatives (Limit RM9000)= RM5000
Amount of education fee for self (Limit RM7000)= RM8000
Amount of equipment for disabled person (Limit RM6000)= RM4000
Amount of medical expenses for self, spouse or children (Limit RM10000)= RM7000
Amount of medical expenses for parent (Limit RM8000)= RM5000

Please enter the details of TAX REBATE :
Zakat & fitrah = RM250

Please enter the Monthly Tax Deduction (MTD) : RM140

----- SUMMARY -----

Yearly Income    = RM40900.00
Total Relief     = RM28000.00
Taxable Income   = RM12900.00
Income Tax       = RM79.00
Rebate           = RM650.00
Total MTD        = RM1680.00

-----

Amount need to return by LHDN = RM1680.00
```

Screen 5: User Page

Screen 5: User has to enter some information for income tax calculation, such as annual income, tax relief, tax rebate, mtd and others. After enter all of the information, the system will generate the summary for the user and conclude that the amount needs to pay back or need to be paid.

```
Thank you for using our system !!
Goodbye...
```

Screen 6: Exit Page

Screen 6: While pressing value 3, the system will be quit.

PART 4: Appendix

Useradmin.h header file code

```
useradmin.h  ×
1  #ifndef USERADMIN_H
2  #define USERADMIN_H
3
4  #include <iostream>
5  #include <string>
6  #include <iomanip>
7
8  using namespace std;
9
10 class Admin;
11
12 class User {
13 private:
14     double yearlyIncome, relief, taxableIncome, incomeTax, rebate, MTD;
15
16 public:
17     User(double yI = 0.0, double rI = 0.0, double tI = 0.0, double iT = 0.0, double rb = 0.0, double mtd = 0.0);
18
19     void setYearlyIncome(double yI);
20     void setRelief(double rI);
21     void setTaxableIncome(double tI);
22     void setIncomeTax(double iT);
23     void setRebate(double rb);
24     void setMtd(double mtd);
25
26     double calcYearlyIncome();
27     double calcTotalRelief();
28     double calcTaxableIncome(double y, double r);
29     double calcIncomeTax(double tI, Admin& objI);
30     double calcRebate(double tI);
31     void calcFinal(double r, double iT, double mtd);
32     void displaySummary();
33
34     friend class Admin;
35 };
36
37 class Admin {
38 private:
39     int id = 007;
40     double rate[12] = {0.00, 0.01, 0.03, 0.06, 0.14, 0.21, 0.24, 0.245, 0.25, 0.26, 0.28, 0.30};
41     double amount[12] = {0.00, 0.00, 150.00, 600.00, 1800.00, 4600.00, 10900.00, 46900.00, 83650.00, 133650.00, 237650.00, 517650.00};
42
43 public:
44     int getId() { return id; }
45     void updateRate();
46     void displayRate();
47
48     friend class User;
49 };
50
51 #endif
```

Main.cpp code

```
main.cpp
1 #include "UserAdmin.cpp"
2 #include <iostream>
3 #include <iomanip>
4
5 using namespace std;
6 int main() {
7     double yI, tR, tI, iT, r, mtd;
8     int choice, temp_id;
9
10    User u;
11    Admin a;
12
13    do {
14        cout << "\n----- Welcome to Income Tax Calculation System ----- \n" << endl;
15        cout << "1. Admin" << endl;
16        cout << "2. User" << endl;
17        cout << "3. Exit" << endl;
18        cout << "\nPlease enter your choice (1/2/3): ";
19        cin >> choice;
20        cout << endl;
21    } while (choice < 1 || choice > 3);
22
23    if (choice == 1) {
24        cout << "\n----- Welcome to the Admin Page -----" << endl << endl;
25        do {
26            int temp_choice;
27            cout << "Please enter your admin ID : ";
28            cin >> temp_id;
29
30            if (temp_id == a.getId()) {
31                cout << "Welcome!" << endl << endl;
32                cout << "----- Current Rate ----- \n" << endl;
33                a.displayRate();
34                a.updateRate();
35                cout << "-----" << endl;
36
37                cout << "\nDo you want to test the income tax calculation system ?" << endl;
38                cout << "1. Yes\n2. No" << endl;
39                cout << "\nEnter your choice : ";
40                cin >> temp_choice;
41
42                if (temp_choice == 1) {
43                    cout << "\n----- Welcome to the User Page ----- \n" << endl;
44                    yI = u.calcYearlyIncome();
45                    u.setYearlyIncome(yI);
46                    tR = u.calcTotalRelief();
47                    u.setRelief(tR);
48                    tI = u.calcTaxableIncome(yI, tR);
49                    u.setTaxableIncome(tI);
50                    iT = u.calcIncomeTax(tI, a);
51                    u.setIncomeTax(iT);
52                    r = u.calcRebate(tI);
53                    u.setRebate(r);
54                    cout << "\nPlease enter the Monthly Tax Deduction (MTD) : RM";
55                    cin >> mtd;
56                    mtd *= 12;
57                    u.setMtd(mtd);
58                    u.displaySummary();
59                } else {
60                    cout << "Logging out from Admin Page....";
61                }
62            } else {
63                cout << "Incorrect ID..." << endl << endl;
64            }
65        } while (temp_id != a.getId());
66    } else if (choice == 2) {
67        cout << "\n----- Welcome to the User Page ----- \n" << endl;
68        yI = u.calcYearlyIncome();
69        u.setYearlyIncome(yI);
70        tR = u.calcTotalRelief();
71        u.setRelief(tR);
72        tI = u.calcTaxableIncome(yI, tR);
73        u.setTaxableIncome(tI);
74        iT = u.calcIncomeTax(tI, a);
75        u.setIncomeTax(iT);
76        r = u.calcRebate(tI);
77        u.setRebate(r);
78        cout << "\nPlease enter the Monthly Tax Deduction (MTD) : RM";
79        cin >> mtd;
80        mtd *= 12;
81        u.setMtd(mtd);
82        u.displaySummary();
83    } else {
84        cout << "Thank you for using our system !!" << endl;
85        cout << "Goodbye...";
86    }
87
88    return 0;
89 }
```

useradmin.cpp code

useradmin.cpp

```
1  #include "UserAdmin.h"
2  #include<iostream>
3  #include <iomanip>
4
5  using namespace std;
6
7  // User Constructor
8  User::User(double yI, double rI, double tI, double iT, double rb, double mtd) {
9      yearlyIncome = yI;
10     relief = rI;
11     taxableIncome = tI;
12     incomeTax = iT;
13     rebate = rb;
14     MTD = mtd;
15 }
16
17 // Mutator Functions
18 void User::setYearlyIncome(double yI) { yearlyIncome = yI; }
19 void User::setRelief(double rI) { relief = rI; }
20 void User::setTaxableIncome(double tI) { taxableIncome = tI; }
21 void User::setIncomeTax(double iT) { incomeTax = iT; }
22 void User::setRebate(double rb) { rebate = rb; }
23 void User::setMtd(double mtd) { MTD = mtd; }
24
25 // User Member Functions
26 double User::calcYearlyIncome() {
27     double act_income, pas_income;
28     cout << "Please enter the details of your ANNUAL INCOME : " << endl;
29     cout << "Active income = RM";
30     cin >> act_income;
31     cout << "Passive income = RM";
32     cin >> pas_income;
33     return act_income + pas_income;
34 },
35
36 double User::calcTotalRelief() {
37     double r = 0.0;
38     double IDR, edu, disabled, m1, m2;
39     cout << "\nPlease enter the details of TAX RELIEF : " << endl;
40     cout << "Amount of individual & dependent relatives (Limit RM9000)= RM";
41     cin >> IDR;
42     if (IDR > 9000) { IDR = 9000; }
43     cout << "Amount of education fee for self (Limit RM7000)= RM";
44     cin >> edu;
45     if (edu > 7000) { edu = 7000; }
46     cout << "Amount of equipment for disabled person (Limit RM6000)= RM";
47     cin >> disabled;
48     if (disabled > 6000) { disabled = 6000; }
49     cout << "Amount of medical expenses for self, spouse or children (Limit RM10000)= RM";
50     cin >> m1;
51     if (m1 > 10000) { m1 = 10000; }
52     cout << "Amount of medical expenses for parent (Limit RM8000)= RM";
53     cin >> m2;
54     if (m2 > 8000) { m2 = 8000; }
55     r = IDR + edu + disabled + m1 + m2;
56     return r;
57 }
58
59 double User::calcTaxableIncome(double y, double r) {
60     return y - r;
61 }
62
```

```

63 double User::calcIncomeTax(double tI, Admin& obj1) {
64     double iT = 0.00;
65     if (tI <= 5000.00) {
66         iT = 0.00;
67     } else if (tI <= 20000.00) {
68         iT = 0.00;
69         tI -= 5000.00;
70         iT += tI * obj1.rate[1];
71     } else if (tI <= 35000.00) {
72         iT = obj1.amount[2];
73         tI -= 20000.00;
74         iT += tI * obj1.rate[2];
75     } else if (tI <= 50000.00) {
76         iT = obj1.amount[3];
77         tI -= 35000.00;
78         iT += tI * obj1.rate[3];
79     } else if (tI <= 70000.00) {
80         iT = obj1.amount[4];
81         tI -= 50000.00;
82         iT += tI * obj1.rate[4];
83     } else if (tI <= 100000.00) {
84         iT = obj1.amount[5];
85         tI -= 70000.00;
86         iT += tI * obj1.rate[5];
87     } else if (tI <= 250000.00) {
88         iT = obj1.amount[6];
89         tI -= 100000.00;
90         iT += tI * obj1.rate[6];
91     } else if (tI <= 400000.00) {
92         iT = obj1.amount[7];
93         tI -= 250000.00;
94         iT += tI * obj1.rate[7];
95     } else if (tI <= 600000.00) {
96         iT = obj1.amount[8];
97         tI -= 400000.00;
98         iT += tI * obj1.rate[8];
99     } else if (tI <= 1000000.00) {
100        iT = obj1.amount[9];
101        tI -= 600000.00;
102        iT += tI * obj1.rate[9];
103    } else if (tI <= 2000000.00) {
104        iT = obj1.amount[10];
105        tI -= 1000000.00;
106        iT += tI * obj1.rate[10];
107    } else {
108        iT = obj1.amount[11];
109        tI -= 2000000.00;
110        iT += tI * obj1.rate[11];
111    }
112    return iT;
113 }

```

```

115 double User::calcRebate(double tI) {
116     double zakat, rebate;
117     double taxRebate = 0.0;
118     cout << "\nPlease enter the details of TAX REBATE : " << endl;
119     cout << "Zakat & fitrah = RM";
120     cin >> zakat;
121     if (tI <= 35000.00) {
122         rebate = 400.00;
123     } else {
124         rebate = 0.00;
125     }
126     taxRebate = zakat + rebate;
127     return taxRebate;
128 }
129
130 void User::calcFinal(double r, double iT, double mtd) {
131     double final;
132     double temp_iT = iT - r;
133     cout << "\n-----" << endl;
134     if (temp_iT < 0) {
135         temp_iT = 0;
136     }
137     if (temp_iT > mtd) {
138         final = temp_iT - mtd;
139         cout << "\nAmount need to pay to LHDN = RM" << final << endl;
140     } else {
141         final = mtd - temp_iT;
142         cout << "\nAmount need to return by LHDN = RM" << final << endl;
143     }
144 }
145
146 void User::displaySummary() {
147     cout << "\n----- SUMMARY -----" << endl;
148     cout << "\nYearly Income = RM" << fixed << setprecision(2) << yearlyIncome << endl;
149     cout << "Total Relief = RM" << relief << endl;
150     cout << "Taxable Income = RM" << taxableIncome << endl;
151     cout << "Income Tax = RM" << incomeTax << endl;
152     cout << "Rebate = RM" << rebate << endl;
153     cout << "Total MTD = RM" << MTD << endl;
154     calcFinal(rebate, incomeTax, MTD);
155 }
156
157 // Admin Member Functions
158 void Admin::updateRate() {
159     double newRate[12] = {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0};
160     cout << "\n-----" << endl << endl;
161     cout << "Please enter the NEW income tax rate" << endl << endl;
162     for (int i = 0; i < 12; i++) {
163         cout << "Rate " << i + 1 << " : ";
164         cin >> newRate[i];
165         rate[i] = newRate[i];
166     }
167     cout << endl << "-----" << endl;
168     cout << "\nTax rate renew successfully!\n" << endl;
169 }
170
171 void Admin::displayRate() {
172     for (int i = 0; i < 12; i++) {
173         cout << "Rate " << i + 1 << " : " << fixed << setprecision(3) << rate[i] << endl;
174     }
175 }

```